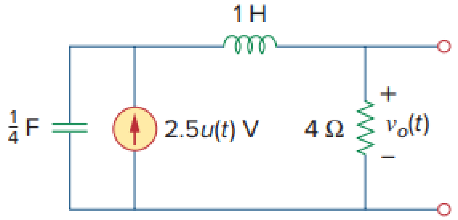
Problem

Determine *vo*(*t*) in the circuit of Fig. 16.6, assuming zero initial conditions.

Figure 16.6:



Step-by-step solution

Step 1 of 4

Refer to the circuit in Figure 16.6 in the textbook.

Transform the circuit to the *s*-domain.

$$2.5u(t) \Rightarrow \frac{2.5}{s}$$
$$\frac{1}{4}F \Rightarrow \frac{1}{sC} = \frac{4}{s}$$
$$1H \Rightarrow sL = s$$

Draw the modified circuit as shown in Figure 1.

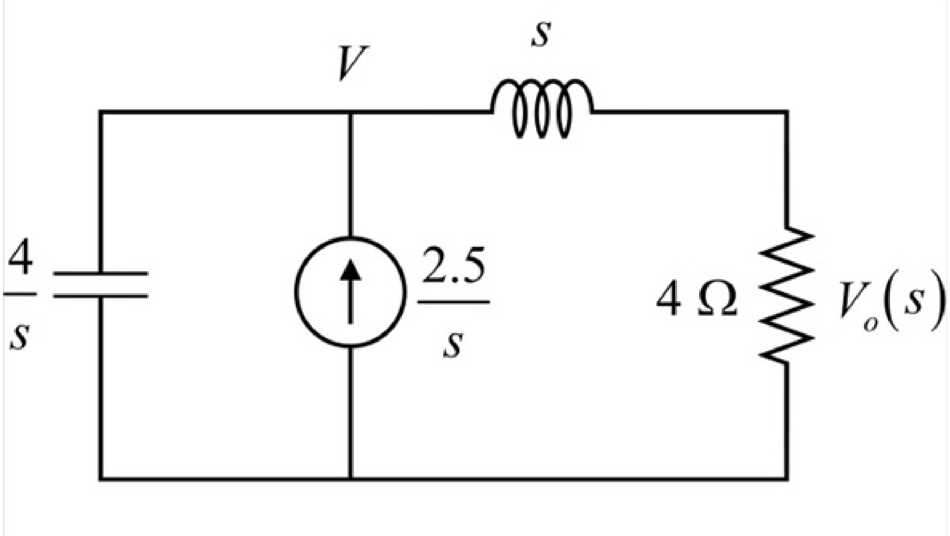


Figure 1

Step 2 of 4

Apply current division principle and ohm's law to calculate the voltage $V_o(s)$.

$$V_o(s)=\left[\frac{2.5}{s}\times\frac{\frac{4}{s}}{\left(\frac{4}{s}+s+4\right)}\right](4)$$
$$=\frac{40s}{s^2\left(s^2+4s+4\right)}$$
$$=\frac{40}{s\left(s^2+4s+4\right)}$$
$$=\frac{40}{s(s+2)^2}$$

$$V_o(s)=\frac{A}{S}+\frac{B}{S+2}+\frac{C}{(S+2)^2} \quad \ldots\ldots (1)$$

$$40=A(S+2)^2+BS(S+2)+CS$$
$$A\left(S^2+4S+4\right)+B\left(S^2+2S\right)+CS=40$$
$$S^2(A+B)+S(4A+2B+C)+4A=40$$

Step 3 of 4

By comparing constant coefficients,

$$4A=40$$
$$A=10$$

By comparing s^2 coefficients,

$$A+B=0$$
$$B=-10$$

By comparing s coefficients,

$$4A+2B+C=0$$
$$C=-20$$

Step 4 of 4

Substitute 10 for *A*, -10 for *B*, and -20 for *C* in equation (1).

$$V_o(s)=\frac{10}{S}-\frac{10}{S+2}-\frac{20}{(S+2)^2}$$

Apply inverse Laplace transform.

$$v_o(t)=10u(t)-10e^{-2t}u(t)-20te^{-2t}u(t)$$
$$=\left[10-10e^{-2t}-20te^{-2t}\right]u(t)\text{ V}$$

Thus, the voltage $v_o(t)$ is $\boxed{10\left[1-e^{-2t}-2te^{-2t}\right]u(t)\text{ V}}$.